

Advanced (Carolina Reaper)

- Q1.** Factor out the greatest common factor from $2x + 6$
(A) $2(x + 3)$
(B) $2(x - 3)$
(C) $3(x + 2)$
(D) $6(x + 1)$
(E) $x(2 + 6)$
- Q2.** Simplify the expression $9x^2 + 12x$ by factoring
(A) $3x(3x + 4)$
(B) $3x(3x - 4)$
(C) $9x(x + 2)$
(D) $12x(x + 1)$
(E) $3x^2(3 + 4)$
- Q3.** Tom has 15 *tospendonvideogames* and 12 on snacks at a gaming store. What is the greatest common factor of the amounts Tom can spend?
(A) 1
(A) 3
(A) 5
(A) 6
(A) 12
- Q4.** A book costs 24 *andapencosts* 8. What is the greatest common factor of the costs?
(A) 1
(A) 2
(A) 4
(A) 6
(A) 8
- Q5.** Factor out the greatest common factor from $18y - 24$
(A) $6(3y - 4)$
(B) $6(3y + 4)$
(C) $12(1.5y - 2)$
(D) $2(9y - 12)$
(E) $3(6y - 8)$
- Q6.** A group of friends want to share some candy equally. If they have 48 pieces and there are 8 friends, how many pieces will each friend get if they factor out the greatest common factor of the total number of pieces and the number of friends?
(A) 4
(B) 6
(C) 8
(D) 12
(E) 16
- Q7.** Simplify the expression $16x^2 + 20x$ by factoring
(A) $4x(4x + 5)$
(B) $4x(4x - 5)$
(C) $8x(2x + 5)$
(D) $5x(3.2x + 4)$
(E) $2x(8x + 10)$
- Q8.** A bakery sells 12 cupcakes and 15 muffins. What is the greatest common factor of the number of cupcakes and muffins sold?
(A) 1
(B) 3
(C) 5
(D) 6
(E) 12

Open Ended Questions (FRQ)

- Q9.** Simplify the expression $25x^2 + 30x$ by factoring and show your work. Assume x is a positive integer.
- Q10.** A store has 36 boxes of pens and 24 boxes of pencils to pack into carts that hold 12 boxes each. What is the greatest common factor of the number of boxes of pens and pencils, and how many carts can be filled with the boxes of pens and the boxes of pencils separately? Show your work and calculations.